Preconditions

`failure_order` shall not be `std::memory_order_release` or `std::memory_order_acq_rel`.

Effects

Atomically compares `expected` to the value stored in `*this` using bitwise comparison and stores `new_value` in `*this` if equal. If the values aren't equal or the update can't be done atomically, updates `expected` to the value read.

Returns

`true` if the existing value of `*this` was equal to `expected` and `new_value` was successfully stored in `*this, false` otherwise.

Throws

Nothing.

Note

The three-parameter overload is equivalent to the four-parameter overload with `success_order==order` and `failure_order==order`, except that if `order` is `std::memory_order_acq_rel`, then `failure_order` is `std::memory_order_acquire`, and if `order` is `std::memory_order_release`, then `failure_order` is `std::memory_order_relaxed`.

Note